

THE TIME LOCK

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Dedicated to my all $\text{\LaTeX}$ users.

THE TIME LOCK

Dr. Angela Rae was head of Space Folding Research at *The Periphery*, a top secret Federation research facility located on the surface of a small moon at the outer fringe of the Kalix system.

Dr. Rae and her 20-something-sized team of researchers were working as fast as they possibly could.

A single thought racing through everyone's mind...

How did they find us?

Up there, orbiting at a chill 26,000 miles, was death. Having covertly decelerated in the radar shadow of a nearby moon, a tiny Rebel ship had suddenly emerged less than an hour ago - catching every living soul on this outer-rim secret research facility by surprise, and spreading fear among its few inhabitants.

Wasting no time, the rebels immediately communicated their terms: *stand down all defences, halt all interstellar communication, and prepare to be boarded.*

With the hostile rebellion force in orbit, and a likely

extermination of all Federation members of this facility, Dr. Rae and her team were frantically gathering all the key discoveries of advanced Space Folding technology that she and the team had made over the last decade. Realizing the danger of having ground-breaking space-faring technology end up in enemy hands, Angela Rae decided to initiate *Facility End Of Life Protocol 2B: "Time-Lock Puzzle"*. The protocol instructed Angela to encrypt all research findings into a time capsule located in the underground vault below the facility.

How did they find us? The thought was stabbing at the back of Angela's mind, while she frantically navigated authorization steps of top secret subprocesses. *This moon isn't even registered in the Federation Naval Charts - it doesn't exist. Do we have a spy on our team?*

Allowing the terrifying thought to linger in her mind, Angela faced the group of researchers gathered around her in the Periphery's Command Central - a grandiose name for a small room littered with computers.

"Everyone, listen up! There is zero chance that Federation forces will reach us in time - we're on our own. None of us expected this, and I have as many questions as you do. In accordance with Federation Regulation, I have decided to initiate the creation of a Timed-Release Cryptographic Capsule. We need to hurry and prime its payload with all essential blueprints, *the Ingegard* algorithm, and confirming test data. If we don't get the capsule ready and locked within the next hour then everything we've worked for risk falling into the wrong hands. That could mean the end of the Federation as we know it. I understand this isn't the way any of us expected things to end. I want to extend my

sincere gratitude to each and every one of you.”

Angela hesitated for a moment, but then decided to follow dictum, and with a straight face exclaimed.

“The Federation thanks you for your sacrifice.”.

The research compute room fell silent, its heavenly-white focus-friendly interior decoration creating a mockery of the grim situation the team was facing. The faint sound of busy compute-hardware carelessly whirring in the background sent chills through Angela’s spine. Her last statement clearly drove home the message, *no one in this room will be alive in a couple of hours*.

The team’s predicament, Angela realised, was the delicate challenge of keeping top secret research out of Rebel hands, until Federation forces return on their scheduled re-supply mission. On the flip side, locking the research away for an indefinite time would place it too far out of Federation scientist’s reach, in a war that the Federation is currently losing.

What was needed, which some Federation geniuses on Earth had realised upon building the the station, was technology to lock away invaluable research secrets for just enough time until Federation forces would return and retake control of the facility.

What was needed, was to send a message into the future.

Dr. Angela Rae, and her team of fellow researchers, had quickly collected all essential research data related to their groundbreaking *Space Folding* innovation. Blueprints of machines, *the Ingegard* algorithm, and confirming test data. An assistant checked the total bitsize of the precious

data;

"Dr. Rae - we've wiped all non-essential data from the mainframe, we're down to just three data files containing project-essential information, for a total of 886 kb of message data."

Angela raised an eyebrow.

"That's all? A decade's worth of research in less than a megabyte?"

The assistant looked around nervously, then returned to face the penetrating gaze of a woman who was infamous for ferreting out mistakes - and ending careers. He shrugged, then started, with a panicky voice. . .

"I mean, it's all the data that Federation scientists will need to recreate the space folding machine without our help, considering we'll all be —"

"FINE!", Angela cut him off.

"Let's move on. We need to calculate the duration of the Time-Lock. Remember, we want to aim for a future date that is close to the expected return and facility-take-over by Federation forces. Go!"

A tense ambience filled the room, as researchers scrambled to save the legacy of their life for the future of the Federation. This went on for a few minutes, until something shifted - one of Dr. Rae's assistants, Embla Garpe - an introvert, dark-haired mathematician, was slowly but surely attracting the attention of everyone in the facility command room.

To Angela and her crew's utter amazement, and against every facility protocol in the entire Federation, Embla . . . was smoking . . . a nicotine cigarette. For a brief moment, everyone just stared at the young woman - white

smoke slowly shrouding her in ethereal mysticism. Equally amazing was the fact that Embla didn't seem to be aware of it. The slender woman was entirely focused on the delicate task of calculating time-lock duration estimations, browsing through Federation re-supply frequency logs, and looking up solar system outlier event probabilities.

If it wasn't for the fact, that death was pretty much knocking on the research facility's bulk door, Angela would curse out the young assistant, instruct security to throw her in the brig, and have her shipped out on the next rotation. Instead, she smiled, and calmly asked Embla . .

"Do you have an ETA of the next Federation ship?".

Without turning in her chair, Embla took a deep drag on the cigarette, which was quickly reaching its own end of life. The moment seemed frozen in time, a slow irreversible stream of entropy particles drifting through the room.

"I do.".

By now, the team's focus had completely switched from Embla to the cigarette in her hand, which was close to burning her fingers. As if on queue, she put out the cigarette (to Angela's horror *On the fucking terminal desk!*), and with complete disregard for Facility Code of Conduct, flicked away the bud into the far corner. Embla leaned back in her chair, staring into a distant future she would never get to experience, and then said. . .

"Based on the Poisson distribution and Principal Component Analysis of all available time-frequency logs, the limit of the log-linear converges with a high confidence and acceptable error margin on the value, 8.02999". She finally turns to face Angela and the rest of the team . . .

“In other words, a Federation ship will arrive . . . in 8 days.”.

In her head, Dr. Angela Rae revised the theory and *raison-d'être* behind time-capsule technology. Over the last seven decades, the Federation had been engaged in a war against the Rebellion. Stretching accross multiple Federation controlled solar systems, the war was slowly being lost to the Rebellion. Having identified the sheer size of the Federation, as the key weakness in its war against an enemy, that could appear-strike-and disappear before Federal forces could engage them - the decision was made to take a high risk.

In a desparate attempt to swing the tides of war, the Federation started investing unprecedented resources into high-risk military research programs, tasked with delivering groundbreaking communication and transportation technology that would enable Federation to enagage the elusive Rebel forces where ever they appeared.

Considering the extreme dangers, and extreme value, of such advanced technology - Fedration Generals found it prudent not to conduct this reaseach in close proximity of highly populated Federation planets. The solution was to hide these secret research facilities at the outer reaches of Federation space. The Periphery, was one such facility, and it had delivered in spade - inventing a groundbreaking new transportation technology that could fold space. In theory, this should enable Federation ships to travel at unprecedented speed, and, according to every battle simulation run so far, swing the tide of war in favor of the Federation.

Given the extreme value of any breakthrough technol-

ogy, the decision was made that, in the event of discovery by Rebellion forces, any breakthrough research was ordered to be saved and locked away in a Capsule - in itself, one of the most advanced pieces of technology in the Federation's arsenal.

Fearing that advanced weaponry would fall into the hands of Rebels, Federation scientist managed to invent a groundbreaking technology - The Capsule. was technology that Federation scientist had invented to protect remote research stations, in the unlikely event that Rebellion would find the research - an entity that tried to *solve the puzzle* contained inside the Capsule, was denoted *the attacker*. Solving the puzzle required the attacker to perform a specific number of computing operations, specifically *squaring a number* until a secret cipherkey was found. The attacker in this case was the *compute hardware of the Rebellion ship*, currently in orbit above Angela Rae's research facility.

An excited voice cut through Angela's concentration.

"Long-range optics confirm it. That's a Class IV Blüm Blüm Shub - an old rebel scout ship." still behind a wall of computer screens . . . "We'll simply have to pray that it hasn't been refitted with a Quantum Shor Solver". Test engineer Tania Teigen popped up with a cynical smile. "Because if they have one onboard, then our cute little capsule won't last 1 quecto second."

Having identified the ship's type, the team quickly cross-reference the research facility's military knowledge base,

"The Rebel ship-computer should have a computing power of" it was Yuyeh, Angela's longtime friend, a brilliant

researcher “presuming these knowledge base is correct, of around 3 million squarings per second.” Yuhey, looked around - hoping for a reaction from the team. “. . . and that’s assuming the rebels manage to allocate 90% of the ship’s compute power to the decryption of the Time-Lock Puzzle.” Not sure if anyone was understanding what he was even talking about, he felt the need to lighten the mood. “Meaning, it’s going to get cold as hell in that ugly ass ship!”, Yuyeh beamed with a slight sadistic grin.

“I know I’m not the genius around here, but—”.

It was Jenz, a member of the Periphery’s Residential Support Team, and generally considered to be the *least* valuable member of the team. There was even a rumor that Jenz father was a big shot Federation General who had pulled some strings to get his son deployed far away from battle hot zones.

“How do we know the Rebels won’t simply steal the capsule, load it onboard their ship, and make a run for it? What am I missing here?”

Angela Rae looked up from the delicate calculations in front of her. She forced a smile. In a few hours she, and everyone else in this remote research facility, would likely be dead or captives on a Rebel ship. The chances of anyone ever seeing their family or loved ones were basically zero. And despite this grim state of affairs, her team were actively trying to solve the problem.

“No, it’s a fair question, Jenz.” Angela started with a weary, but forgiving tone.

“As most of you know, the top secret research we’ve done here is considered by the Federation to be a top priority. In the creation of this site, no cost was considered

too high, no resources were spared. The Capsule, located 3 kilometers below this room, is protected by a state of the art military force field. There's no technology, that we know of, capable of displacing the Capsule even one milimeter from its locus. Not even a nuclear strike. The Rebels can interact with it, but it can't be physically moved. You can think of it, as a pretty good safe.”

Angela returned to the work at hand, fighting back tears. She checked the time, made a quick risk calculation, consulted her training on tactical decision theory in siege-like scenarios, and reached a conclusion.

“Listen up, everyone. . . .please!” Angela surved each of the members of her team. “These estimates will have to do - we’re simply out of time. We’re expecting Rebel fighters to land on this site in less than an hour and breach the facility. We must proceed to the next phase, and start priming the Capsule payload.”

A loud “PING!” startled Angela, and everyone else in the command center. A proximity warning informed them that Kalman-tracking-procedures had detected an inbound non-Federation-registered vessel - a highly unlawful and equally distressing event.

“Oh, God! The Rebels are almost here. Quick, someone fetch me a Vanguard suit and help me put it on. I have to get down to the Capsule in the vault, and start the Time-Lock procedure.”

Angela looked at her team, wondering if they still respected her authority.

“On my way.”, Yuyeh shouted as he scrambled through the room to go and get a Vanguard hazardous environment

suit, capable of withstanding the -90°C temperature inside the vault.

Having quickly suited up in the Vanguard, surrounded by teary-eyed colleagues doing their best not to panic and breakdown crying, Angela Rae was now at the tail-end of the the 6-minute elevator ride down to the vault, which housed the Capsule. Breathing in the Vanguard suite bothered her, but given the circumstances she was past caring.

On passing 2.7 km below ground level, communication with the topside command center had suddenly been cut off. She didn't know if it was due to Rebels finally cutting through blast doors and killing everyone, or if top side radio signals were simply unable to reach her at this depth. Regardless, no one topside had responded to her last couple of half-hearted hails. She felt a stinging sense of guilt having left her team - funny enough, riding down in the elevator felt like the opposite of a captain going down with the ship.

The elevator decelerated to a complete standstill. A voice-assisted control panel on Angela's right demanded that she accept a full biometric authorization scan. She complied. "Scanning. . . . Welcome Dr. Rae." The female voice was provokingly calm, and as the elevator door opened, Angela couldn't help but curse the synthetic voice from the inside her suit.

"Bitch, I'm about to die, okay? Nevermind, don't answer that." She reminded herself the job she was there to carry out. She reminded herself, that, at this very moment, members of her team might be getting killed or tortured by Rebels, hoping to gain access to the capsule.

As Angela Rae stepped out of the elevator she felt unease - The Vault was much smaller than she had imagined. She'd been briefed about it, but that was over ten years ago. However, there was one thing that aligned well with the fragments of memories that she still had about this place. Some fifty meters straight ahead of her was a massive construct, and in the middle of it floated a dark orb-like . . . "thing" - The Capsule.

As she approached the Capsule it seemed to react to her presence. There was a humming sound, some kind of vibration in the air - *The force field?* The Capsule was beautiful, and ugly, at the same time. A mix of organic and synthetic technology.

Suddenly a request for a direct neural link was made . . . from the Capsule it seemed.

She recognized the procedure of neural link communication, from the training she had received as a young cadet at the Federation Military Research Academy.

She carefully provided the authorization *thoughts*, uneasy of what would happen next. Suddenly, a tremendous feeling of power flowed through Angela, as she gains direct control of the Capsule.

Angela was exhausted. She had lost count of the number of physical concern warnings that the Vanguard suite had flashed in the center of her visor. Right now, she needed to focus on the next phase of the Time-Lock Puzzle activation procedure. In this cold and fateful place, the Capsule in front of her was waiting for Angela to provide two pieces of information, *how long* and *how hard*.

Angela spent a moment to remind herself that *how long*

was the approximate number of seconds that the team wanted the Time-Lock Puzzle to withstand the Rebel's decryption efforts. Further, *how hard* was the number of squarings per second that her friend Yuhey had pulled from the research facility's knowledge base, it represented an assumption of the Rebel ship's computational power. The resulting, was the Time-Lock's Puzzle Strength - the total number of squarings that the Rebels would have to perform to *solve* Time-Lock Puzzle.

Next, Angela entered Embla Garpe's estimate of how much time was left until the next Federation re-supply ship would arrive in the Kalix system, and take control of the The Periphery: 691,200 seconds, or 8 days. Finally, Angela entered Tania Teigen's guestimated squaring power of the Rebel ship: 3,000,000.

The Capsule instantly displayed the total number of squarings that an attacker would need to perform to unlock it, once the Time-Lock Puzzle was activated: $691,200 * 3,000,000 = 2,073,600,000,000$.

"Two trillion squarings to solve the Time-Lock Puzzle. Is that really strong enough?" Feelings of anxiety washed over Angela, ". . .what if that pretentious cigarette-smoking bitch, Embla, made an arithmetic mistake and we die for nothing?"

Realising that she was talking to herself, and feeling ashamed of the way she's expressing her doubts over colleagues who were probably dead or dying - Angela squashed the urge to 10x the Puzzle's Strength. She decided to trust her team, and proceeded by carefully entering the values for *how long* and *how hard*, via the Capsule's neural arming interface.

Having assumed that she was the only living organism in the vault, Dr. Angela Rae could barely believe her senses when she heard a sound behind her. She slowly turned around, just in time to see the elevator doors closing. . . in horror she witnessed the elevator accelerate up, towards the topside facility Command Center.

Her scientist mind immediately started modelling the situation, *I'm the only Federation member with a security clearance to access that elevator. The Rebels must have breached our computer systems and somehow circumvented our security systems to gain access to the Vault elevator.* . Angela thought, as her mind was racing into different scenarios, desperately looking for a probable outcome that did *not* land in the one staring right at her. *The elevator ride down took six minutes. That means. . . I have less than 12 minutes to finish the locking procedure before this place is swarming with Rebels.*

Angela instructed her Vanguard suite to set a count-down timer for 12 minutes. In the top right corner of her helmet, numbers started counting down.

The next phase of the Time-Lock Puzzle creation called for Angela to initiate a Capsule procedure that would generate a secret key. That key was of extreme importance since it would later be used to encrypt the teams precious research data - a decades worth of Space Folding research work and a transportation technology that Angela knew would forever change interstellar travel.

The key had to be secure enough so that the Rebels would be unable to break it by brute force. The key also had to assume that advances in computing power technology would be made during the expected lifetime of the Time-Lock Puzzle. Angela reminded herself that nothing

groundbreaking, in terms of computing power, was likely to materialize in the hands of the Rebels over the next 8 days - the estimated time of arrival of the next Federation ship.

The blinking 00:08:37, in the upper right corner of her helmet's information display, informed Angela that she had around eight minutes left to live. She had started the countdown the moment she realized that the Rebels had likely succeeded in gaining access to the elevator, which was the only way to reach the vault.

Having completed all the necessary preparatory steps, Angela now started the transfer of the tiny *886 kb message* payload. Invaluable blueprints of innovative Space Folding technology, was now being transferred from the computer of Angela's Vanguard suite over to the Capsule. She started swearing, impatiently authorizing a long list of security requests regarding the encryption of message data.

"The folks who created this alien bowling bowl sure weren't in a hurry. I'll be dead long before this thing. . ."

A brief a humming sound, followed by the Capsule informing Angela that the message payload, containing blueprints, the Ingegard algorithm, and the confirming test data - had successfully been encrypted and stored inside the Capsule.

With the secret key K generated in the previous step, the Capsule proceeded to encrypt Dr. Angela Rae's fateful message. She anxiously waited for the Capsule to report that the procedure was complete . . . but nothing happened. She suddenly felt the urge to bang on the thing with her clenched fist.

“Hurry! You stupid fucking computerfuck!” Angela screamed at the black orb-like thing, floating in mid-air in front of her. Still, no reaction from the Capsule, and as the minutes passed, Dr Angela Rae began to cry.

“I knew this idea would *never* work.”

Dr. Angela Rae was close to having a mental breakdown. Her Vanguard environment suite was scrambling to issue a truckload of chemicals into her bloodstream, all in an effort to lower her racing heart rate. A 300 milliseconds long stream of positive imagery suddenly flashed directly into Angela’s iris, an effort by the Vanguard to trigger a serotonin response that would offset her accelerating depression.

“Cows on green spring pastures? Really? I’m minutes from having my brain split open by Rebel lasers. . . and some Federation astro-psychiatrist had the idea that images of happy cows will make me calm down?”.

No answer. Not even from the dark mysterious orb floating in midair in front of her. If only she had someone to talk to. “Yuhey?!” , she tried - nothing.

The countdown had just entered the territory of 2 minutes, 00:02:58 to be exact. Leaning on decades of discipline, and soothing images of cows on green pastures, Angela willed herself into action. The cipher key was one of the last elements that needed to be created, in order to activate the Time-Lock Puzzle, lock the Capsule, and declare mission accomplished.

With one minute left of her life, Dr. Angela Rae - Head of a Top Secret Federation advanced research program, located on a remote moonbase no one would ever hear

about - was enjoying the sweet overture of Johann Sebastian Bach's *Orchestral Suite No.3 in D-dur*. The 1,281 year-old masterpiece, was flowing through the loudspeakers of her Vanguard suite, a serenade of Angela's accomplishment.

"One last step, and it's Check Mate on these Rebel!" Angela beamed with pride and an extreme sense of accomplishment.

In front of her, the black alien-like Capsule, was patiently waiting for her to issue the final *Initiate Locking* command, which would seal the Capsule for a duration of 2 trillion squarings.

As a Federation Officer assigned to a military research program, and the only holder of the secret key which could instantly open the Capsule - Angela knew she could not allow the Rebels to capture her alive. They would simply torture her until she told them everything, including the name of her secret lover.

Bypassing a slew of security clearances, she authorized and ordered her Vanguard to stand by for a word command and instructed the suit to, upon the word command, inject fatal doses of a military nerve agent into her bloodstream. Returning to the Capsule, Angela initiated the final step, authorizing the destruction of the secret keys that were used to during the creation of the Time-Lock Puzzle. Without these two secret keys, it would be impossible to open the Time-Lock Capsule. In order to open the Capsule the Rebels would have to decrypt it by performing two trillion squaring operations.

Over the Neural Link, the Capsule confirmed the deletion of the two secret keys. Beaming with the feeling of having beat the Rebels and accomplished the task that she

and her, now presumably dead, team set out to do, Angela ordered the Capsule to the final step.

Commence Locking Procedure and give me a softcopy when the locking procedure is -

Angela stopped mid-thought, something in the corner of her eye was screaming for her attention. As she turned, some 30 meters away from her, the elevator had just finished decelerating to a complete standstill. Doors opening . . .

“Yueh?!” *What was he doing here?*, Angela gasped. She was stunned to see the face of her longtime colleague, but she was shocked to see dark shadows with raised weapons flash towards her from behind Yuhey.

Her military training wrestled her curious research faculty out of the way, as she raised her arms in a fencing posture, and screamed for the Vanguard suite’s computer to deliver the killing toxin into her bloodstream.

With shocking speed, Rebel hands were already upon her, ripping open the Vanguard suite. A sharp blow from an instrument, just below her right kidney. Something cold, spreading.

Chemically induced stasis? Are they trying to put me in a comaaaaa?.

Angela felt two chemicals, Vanguard vs Rebel, fighting each other. One trying to kill her, the other she now realised with horror, trying to prevent her from dying. An unbelievably sharp pain seered her brain.

Screaming, she fell to the ground. the minus -90 degrees celsius of the vault sweeping across her now suitless body, but it barely registered by Angela’s now rapidly failing nervous system. With her exposed retinas failing to

deliver signals that her brain could make sense of, darkness caved in on Dr. Angela Rae. Despite a cocktail of chemicals now spreading throughout her body, she no longer felt pain. Surprisingly, the prominent feeling was that of pride and accomplishment.

A synthetically amplified human voice cut through the darkness . . .

“Sir, we’re losing her. She managed to take a suicide toxin before we got to her. Our nanoserum is keeping her alive, but she’s in a pretty bad shape.”

Another voice, this one older . . . and calmer.

“Sever the blood flow to her brain. Tap into her cortex. Full tissue hardcopy.” The Rebel realising that the whole operation was starting to go sideways.

“Do it!” he snapped.

On the floor, a tiny part of what was once Dr. Angela Rae’s was trapped in limbo. A deadly neurotoxin compelling her to give up, give in, and die. But hardcoded organic reflexes, of what was left of her neural system, was refusing the offer. Something, somewhere, was enticing her to remain . . . and reveal? Reveal what? Her curiosity piqued, the balance shifting towards . . . something.

“Sir, the neurotoxin has been neutralized. Brain tissue corruption is over 85%. I cannot confirm that the keys to the Capsule can be extracted from what’s left of her brain.”

The Rebel Strike Team Commander, standing over what remained of the Federation scientist, was clearly disappointed . . . but hopeful.

“Copy. Transfer her to the Shub and initiate extraction procedures - salvaging that key is of extreme importance. And bring *that guy!*”.

The commander pointed to Yuyeh.

“He might be useful in creating a memory trap, that we can lure her conscious into. You two!, secure the facility, cut off any lines of escape, and set up T1C comms. We might be here longer than we plan-”

Suddenly, a tiny shift in the airwaves around the Rebels.

The Rebel commander swung around, weapon raised at the Capsule.

“What. Just. Happened?”

A tiny status indicator on a nearby panel enlightened him, CAPSULE LOCKING PROCEDURE COMPLETE.

“Fuck!”

The Rebel Commander never expected this mission to be a walk in the Sim Park, but the current situation is best described as a tactical conundrum.

“Our medical team sure know their shit.”

On the table in front of him lies a grainy image print-out of a number. During the last four hours, the team managed to salvage and extract fragments of a few visual memories from the female Federalist’s damaged brain. Most memories were stored in old neuron clusters, encoding what the team interpreted to be childhood memories; learning to ride a jetbike, some kind of 1st-in-class trophy, acceptance to the Federation Juvenile Military Academy, and so forth.

On the verge of giving up, the Rebel medical team then stumbled on a tiny neuro-cluster, bordering a massive volume of dead neurons. Similar to the “childhood memory” cluster, this tiny cluster of neurons appeared to encode the aggregate feeling of “pride and accomplishment”. How-

ever, in contrast, these neurons had recently been stimulated!, not by the medical team's nano robots, but by a surge of pure old-fashioned neurotransmitters, presumably to encode a recent and strong sense of accomplishment by the host.

The medical team then decided to try a new, but highly controversial, Interrogation NeurOp Procedure. They tortured Angela's long-time colleague, Yuhey, and fed the recorded sound of his primal screams straight into her still functional hearing receptors. As expected, these horrific signals triggered a deeply traumatic experience in Angela's half-dead brain. Her brainstem responded by releasing a massive burst of excitatory neurotransmitters. This in turn gave rise to neural activity in the form of new neural pathways being created in almost every area of her brain which still contained living neurons, to encode potentially life-saving information. With this "trap" in place, the nano robots managed to entice individual neurons in the tiny cluster to connect, and once chemically consolidated, an "interface" into the high-value target memories was complete. The NeurOp had succeeded.

Building on top of that breakthrough, the team then relied on 31st century standard medical procedures for patients with traumatic brain injury - to map, copy, and virtualize the structure, density, and impedance of the cluster. Basically creating a biosynthetic 1:1 copy of the memory cluster, which could then be rendered *back* into the format that triggered the memory encoding in the first place. The final result of the entire operation now lay in front of the Rebel commander, in the form of a printed image, depicting a number:

691200691200

On the other side of the table, a female Sargent was the first one to speak.

"We've run a few million Posits on the ship's computer. Provided our context, they all converge on *that* number representing *time*, which its standard unit being *seconds*. The amount corresponds to *8 days*. From there, however, the confidence value in any extrapolation, into how 8 days relates to anything, drops quickly . . . except for one –"

"It's the number of days left," the commander interrupted, "until a military Federation ship will enter this system, hail this base, understand that something is wrong. . . and we'll be knee-deep in shit."

With obvious concern, the commander looked at the handful of senior rebels gathered around the planning table in the center of the Shub's brig.

"In other words, we have 8 days to break into that Capsule."